REB, The Course
Class 13 Learning Activity Solution

Cells are growing in an isothermal, steady-state CSTR. The effective reaction, in mass units, is given in equation (1). The reaction is auto-catalytic, and the rate is accurately described by the Monod equation, equation (2), with V_{max} equal to 0.014 min^{-1} and K_m equal to 0.001 g cm^{-3}. Through an appropriate startup process, the 4.43 L reactor can operate at steady-state with no cell mass in the feed. Specifically, if 60 cm^3 min^{-1} of feed containing only substrate at a concentration of 0.04 g cm^{-3}, the steady-state product stream contains substrate at a concentration of 2.97 x 10^{-2} g cm^{-3} and cell mass at at concentration of 4.68 x 10^{-3} g cm^{-3}. However, reactor operation is parametrically sensitive to the volumetric flow rate. Show that to be true by calculating the transient outlet concentration of cell mass as a function of time if the inlet volumetric flow rate changes from 60 to 61 cm^3 min^{-1}.

$$2.2\,S \rightarrow X \tag{1}$$

$$r_1 = \frac{V_{max} C_S C_X}{K_m + C_S} \tag{2}$$

Assignment Summary

Reaction

$$2.2\,S \rightarrow X \tag{1}$$

Rate Expressions

$$r_1 = \frac{V_{max} C_S C_X}{K_m + C_S} \tag{2}$$

Reactor: Liquid-phase, isothermal, transient

Given Constants: V_{max} = 0.014 min^{-1}, K_m = 0.001 g cm^{-3}, V = 4.43 L, $\dot{V}_{in}$ = 61 cm^3 min^{-1}, $C_{S,in}$ = 0.04 g cm^{-3}, $C_{X,in}$ = 0 g cm^{-3}, $C_{S,0}$ = 2.97 x 10^{-2} g cm^{-3} , and $C_{X,0}$ = 4.68 x 10^{-3} g cm^{-3}.

Deliverables: C_X vs. t as a graph

Completion Strategy

This assignment involves the transient response of a CSTR to a change in the feed flow rate. Instead of a reaction, biological cells are growing in the reactor, so mass concentrations and an effective stoichiometry will be used. I can solve the transient CSTR design equations for the outlet mass flow rates versus time and then calculate the equivalent mass concentrations.

Formulation of the Calculations

CSTR Design Equations:

$$\frac{d\dot{m}_S}{dt} = \frac{\dot{V}}{V}\left(\dot{m}_{S,in} - \dot{m}_S - 2.2Vr_1\right) \tag{3}$$

$$\frac{d\dot{m}_X}{dt} = \frac{\dot{V}}{V}\left(\dot{m}_{X,in} - \dot{m}_X + Vr_1\right) \tag{4}$$

CSTR Model Variables: t, $\dot{m}_S$, and $\dot{m}_X$

Additional Computable Unknowns: $\dot{V}$, $\dot{m}_{S,in}$, r_1, C_S, C_X, and $\dot{m}_{X,in}$

$$\dot{V} = \dot{V}_{in} \tag{5}$$

$$\dot{m}_{i,in} = \dot{V}_{in}C_{i,in} \quad i = S \text{ and } X \tag{6}$$

$$C_i = \frac{\dot{m}_i}{\dot{V}} \quad i = S \text{ and } X \tag{7}$$

IVODE Solver Requirements:

1. Initial values of the CSTR model variables.

$$t\Big|_{t=0} = 0 \tag{8}$$

$$\dot{m}_i\Big|_{t=0} = \dot{V}C_{i,0} \qquad i = S \text{ and } X \tag{9}$$

2. Stopping criterion

$$t = t_f \tag{10}$$

3. Derivatives function that
   a. receives values of t, $\dot{m}_S$, and $\dot{m}_X$ at the start of an integration step
   b. calculates the additional unknowns using equations (2) and (5) - (7)
   c. evaluates and returns the design equation derivatives using equations (3) and (4)

Deliverables Function

1. Call the CSTR model function to get $\underline{t}$ and $\underline{\dot{m}}_X$
2. Calculate the concentration

$$\underline{C}_X = \frac{\underline{\dot{m}}_X}{\dot{V}} \tag{11}$$

3. Plot the results

Implementation of the Calculations

The Python script, activity_13.py, and Matlab function, activity_13.m, that were written to perform the calculations accompany this solution.

Results and Discussion

The calculations were performed as described above. The final time, t_f, was adjusted so that the transient filled the graph. The results are shown in Figure 1.

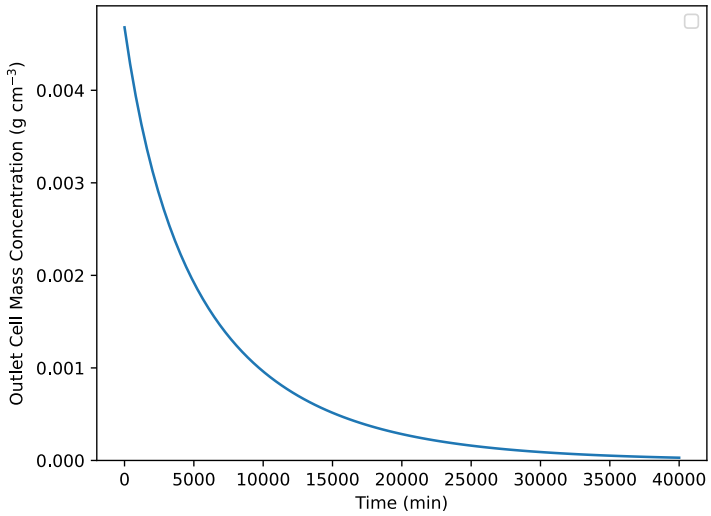


Figure 1. Cell mass washout upon changing the feed rate from 60 to 61 cm^3 min^{-1}

Cell growth is like an autocatalytic chemical reaction, the rate increases when the product concentration (in this case, cell mass) increases. It differs in that most autocatalytic chemical reactions have a small rate when no product is present. In contrast, cells can't grow if there aren't any present.

Indeed, a well-designed startup procedure would be needed to reach the initial steady state in this system. There is no cell mass in the feed. As a result of the startup procedure, the rate of cell mass production within the reactor exactly matches the rate at which cell mass is leaving the reactor.

This can result in a situation with extreme parametric sensitivity. Specifically, If the flow rate through the reactor is increased such that the cell mass leaves the reactor faster than it is generated within it, all of the cell mass eventually will be swept out of the reactor. This phenomenon, sometimes referred to as washout, is illustrated in this example. This reactor was producing cell mass at a steady state, until a small increase in flow rate caused all of the cell mass to be washed out of the reactor.

Summary

The initial steady state was parametrically sensitive to the volumetric flow rate, and when it was increased by less than 2%, from 60 to 61 cm^3 min^{-1}, the reactor contents were washed out.